Quick Sort Algorithm

Merge Sort - Shortcomings

- Merging L and R arrays create a new array
- No obvious way to efficiently merge *<u>in place</u>*
- Extra storage is required to merge and can be costly

Motivation for Quick Sort

- Array A[1..n] can be divided into two parts with the help of the pivot ‘p’ so that everything to the left is smaller than 'p’ everything to the right is bigger than ‘p’.

- No need to Merge

Quick Sort - Introduction

- Tony Hoare - Early 1960's
- Well Known Computer Scientist
- Turing Award Winner

Quick sort - Idea

- Choose a pivot element
- Typically the first/last value in the array is pivot
- Divide/Partition the array A into lower and upper parts w.r.t pivot
- Move pivot between lower and upper partition
- Recursively sort the two partitions

Quick Sort : Divide and Conquer

- **Divide :**
  - Partition the array A[p..r] to two sub arrays A[p..q-1] and A[q+1..r], where q is computed as part of the *PARTITION* function
  - Each element of A[p..q-1] is less than or equal to A[q]
  - Each element of A[q+1..r] is greater than A[q]
  - The sub arrays can be empty or non-empty

Quick Sort : Divide and Conquer

- **Conquer :**
  - Sort the two sub arrays A[p..q-1] and A[q+1..r] by recursive calls to quick sort

Quick Sort - Divide and Conquer

- **Combine :**
  - The two sub arrays are already sorted
  - The entire array A[p..r] is already sorted
  - Nothing particular to do in *combine* step

Pseudocode of Quick Sort

```
QUICKSORT(A, p, r)

if p < r
    then q = PARTITION(A, p, r)
            QUICKSORT(A, p, q-1)
            QUICKSORT(A, q+1, r)
```

Pseudo code : *PARTITION*

```
PARTITION (A, p, r)
x = A[r]
i= p - 1
for j = p to r - 1
        if A[j] ≤ x
                then  i=i+1
                Exchange A[i] with A[j]
Exchange A[i +1] with A[r]
return  i +1
```

How do we partition an array ?

- Suppose array A[1..10] =[9, 7, 5, 11, 12, 2, 14, 3, 10, 6]
- Array entry A[r] is selected as the pivot element, where r is the index of the last element of an array
- A[r] is compared with the array elements until a smaller element *s* (less than or equal to pivot) is obtained
- If *s* is obtained, it is exchanged with the first element of the array initially
- A[r] is again compared with all other elements of A and exchange of smaller element is done further

Working of *PARTITION*

- A[1..10] = [9, 7, 5, 11, 12, 2, 14, 3, 10, 6]
- Pivot = A[r] = 6
- 6 is compared with the elements of A until a smaller element (than 6) is obtained
- In this case, 6 is compared with 9,7 and 5
- Exchange 9 with 5 since $5 \leq 6$
- A = [5, 7, 9, 11, 12, 2, 14, 3, 10, 6]

Working of *PARTITION*

- A = [5, 7, 9, 11, 12, 2, 14, 3, 10, 6]
- Again 6 is compared with 11,12 and 2
- 7 and 2 are exchanged
- A= [5, 2, 9, 11, 12, 7, 14, 3, 10, 6]
- Again 6 is compared with 14 and 3
- 9 and 3 are exchanged
- A= [5, 2, 3, 11, 12, 7, 14, 9, 10, 6]

Working of *PARTITION*

- A= [5, 2, 3, 11, 12, 7, 14, 9, 10, 6]
- Again 6 is compared with 10
- We can see that 5, 2 and 3 are lesser or equal to than the pivot and we have compared pivot with all other elements of A
- Exchange pivot with 11 so that all the elements before pivot is less than or equal to pivot and all the elements after pivot is greater than pivot
- A= [5, 2, 3, 6, 12, 7, 14, 9, 10, 11]
- [5, 2, 3] and [12, 7, 14, 9, 10, 11] are two partitions separated by the pivot 6

Pseudo code of *PARTITION*

```
PARTITION (A, p, r)
x = A[r]
i= p - 1
for j = p to r - 1
        if A[j] ≤ x
                then  i=i+1
                Exchange A[i] with A[j]
Exchange A[i +1] with A[r]
return  i +1
```

Working of *PARTITION*

```
 p                                r
9, 7, 5, 11, 12, 2, 14, 3, 10, 6

 p                                r
9, 7, 5, 11, 12, 2, 14, 3, 10, 6
i j

 p                                r
9, 7, 5, 11, 12, 2, 14, 3, 10, 6
i    j

 p                                r
9, 7, 5, 11, 12, 2, 14, 3, 10, 6
i       j
```

p r

9, 7, 5, 11, 12, 2, 14, 3, 10, 6

i j

r

5, 7, 9, 11, 12, 2, 14, 3, 10, 6

i j

r

5, 7, 9, 11, 12, 2, 14, 3, 10, 6

i j

r

5, 7, 9, 11, 12, 2, 14, 3, 10, 6

i j

r

5, 7, 9, 11, 12, 2, 14, 3, 10, 6

i j

r

5 , 7, 9, 11, 12, 2, 14, 3, 10, 6

i j

r

5 ,2 , 9, 11, 12, 7, 14, 3, 10, 6

i j

r

5 ,2 , 9, 11, 12, 7, 14, 3, 10, 6

i j

r

5 ,2 , 9, 11, 12, 7, 14, 3, 10, 6

i j

r

5, 2, 9, 11, 12, 7, 14, 3, 10, 6

i j

r

5, 2, 3, 11, 12, 7, 14, 9, 10, 6

i j

r

5, 2, 3, 11, 12, 7, 14, 9, 10, 6

i j

r

5, 2, 3, 6, 12, 7, 14, 9, 10, 11

i j

5, 2, 3, 6, 12, 7, 14, 9, 10, 11

5, 2, 3, 6, 12, 7, 14, 9, 10, 11

- [5, 2, 3] and [12, 7, 14, 9, 10, 11] are two partitions separated by the pivot 6

- All the elements before pivot is less than or equal to pivot and all the elements after pivot is greater than pivot

Design of *PARTITION*

- How many counters/pointers do we need?
  - One counter (j) which goes from p to r-1 : to compare all other elements of A with pivot
  - Another counter (i) which keeps track of the position of the smaller element (less than or equal to pivot)

Exercise

- Trace the working of *PARTITION* with the sub array [5,2,3]

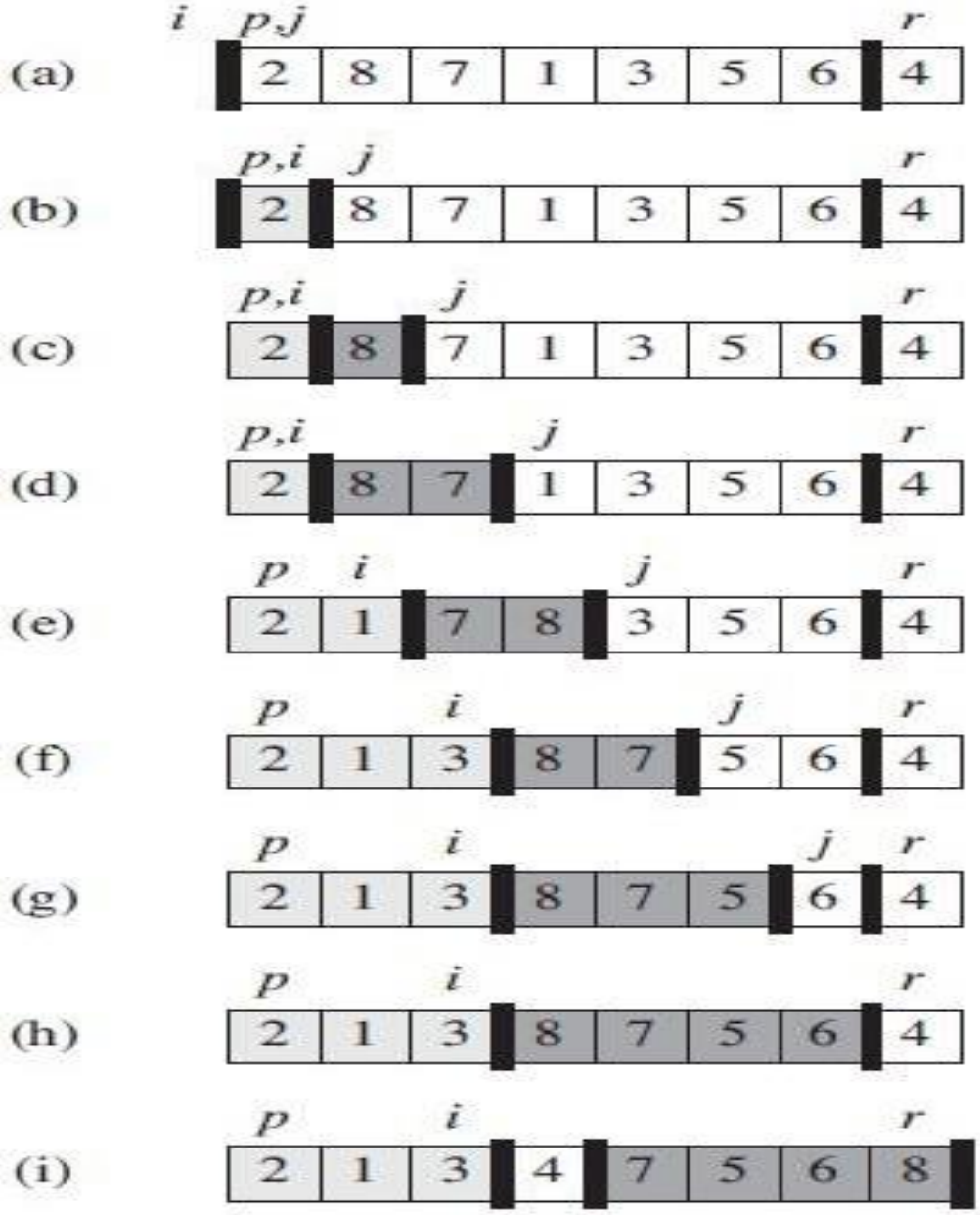
- Trace the working of *PARTITION* with the sub array [12, 7, 14, 9, 10, 11]

- Trace the working of Quick sort with the array [2 4 6 7 9 23]

- Trace the working of Quick sort with the array [26 24 15 7 3 2]

Correctness of *PARTITION*

- PARTITION selects an element x = A[r] as a ***pivot*** element around which to partition the subarray A[p .. r]
- As the procedure runs, it partitions the array into **four** (possibly empty) regions
- Suppose A = [2,8,7,1, 3,5,6,4]


(a) i p,j r
2 8 7 1 3 5 6 4
(b) p,i j r
2 8 7 1 3 5 6 4
(c) p,i j r
2 8 7 1 3 5 6 4
(d) p,i j r
2 8 7 1 3 5 6 4
(e) p i j r
2 1 7 8 3 5 6 4
(f) p i j r
2 1 3 8 7 5 6 4
(g) p i j r
2 1 3 8 7 5 6 4
(h) p i r
2 1 3 8 7 5 6 4
(i) p i r
2 1 3 4 7 5 6 8

Four regions(from fig.(f)

- 1st region : Values no greater than pivot
  (Lightly shaded array elements in previous fig.(f))
- 2nd region : Values greater than pivot
  (Heavily shaded array elements in fig.(f))
- 3rd region: Values not processed for both 1st 2nd regio
  (Unshaded elements from fig.(f))
- 4th region: The pivot A[8] element itself

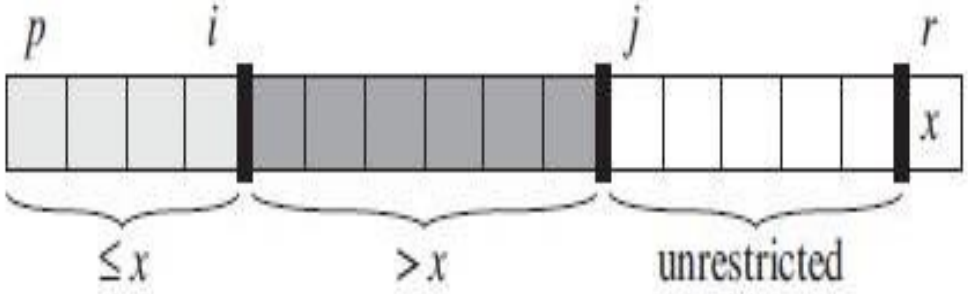


Figure 7.2 The four regions maintained by the procedure PARTITION on a subarray $A[p \,..\, r]$. The values in $A[p \,..\, i]$ are all less than or equal to x, the values in $A[i+1 \,..\, j-1]$ are all greater than x, and $A[r] = x$. The subarray $A[j \,..\, r-1]$ can take on any values.

Loop invariant

- At the start of each iteration of *for* loop in lines 3- 6 , the four regions satisfy certain properties:
- We state those as **loop invariant** :

- At the beginning of each iteration of the loop (lines 3- 6), for any array index *k* :
  - **If** ***p<= k <= i , then A[k] <= x***
  - **If** ***i +1 <= k <= j-1, then A[k] >x***
  - **If** ***k=r, then A[k] = x***

```
PARTITION (A, p, r)
1 x= A[r]
2 i = p - 1
3 for j = p to r - 1
4         if A[j] <= x
5                 i=i+1
6         exchange A[i] with A[j]
7 exchange A[i +1] with A[r]
8 return i +1
```

Observation

- The indices between j and r-1 are not covered by any of the above three cases
- The values in these entries have no particular relationship to the pivot x

Correctness using loop invariant

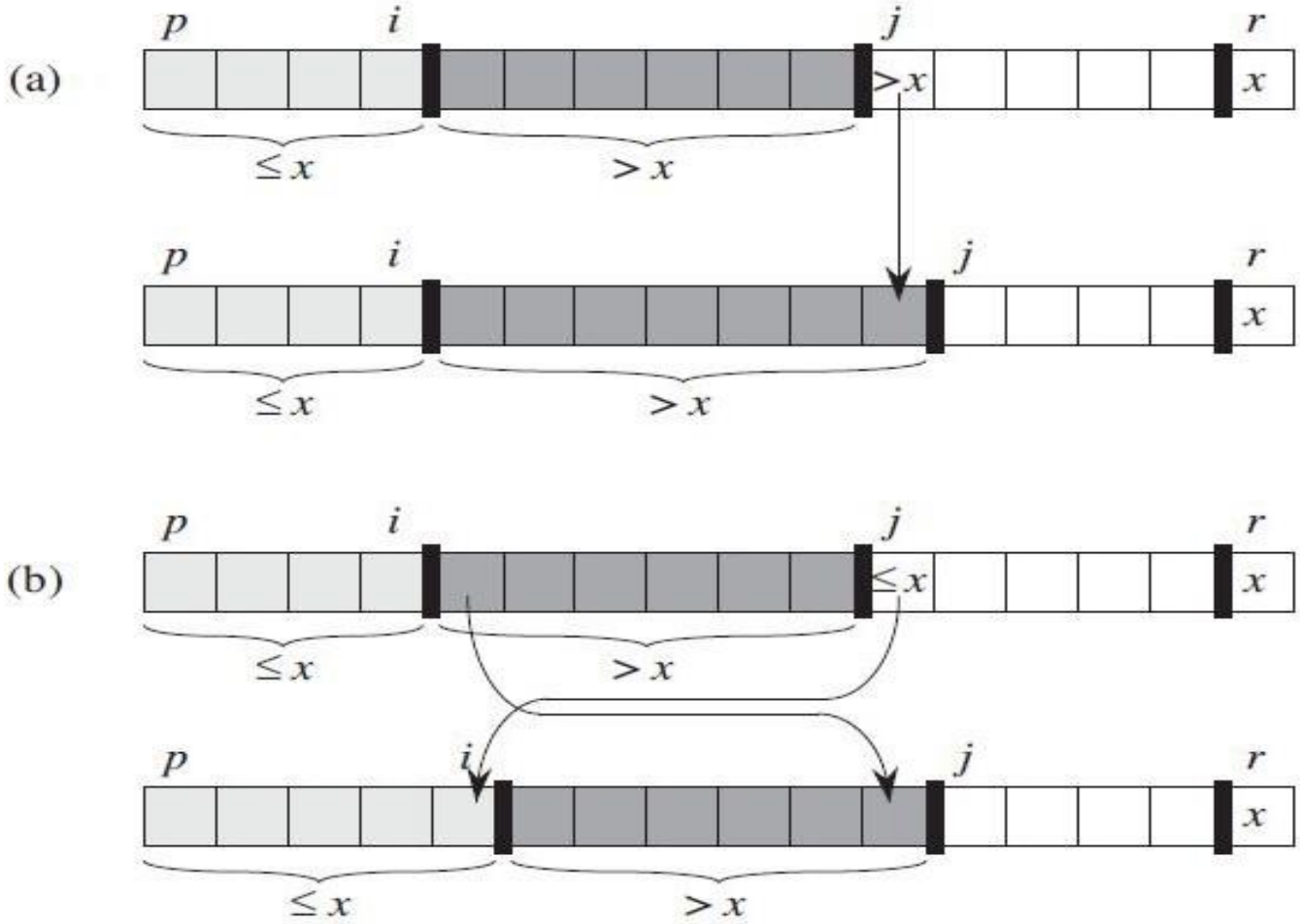
- Loop invariant:
- At the beginning of each iteration of the loop (lines 3- 6), for any array index *k* :
  - **If** ***p<= k <= i , then A[k] <= x***
  - **If** ***i +1 <= k <= j-1, then A[k] >x***
  - **If** ***k=r, then A[k] = x***
- We need to show that:
  - The loop invariant is true prior to the first iteration
  - Each iteration of the loop maintains the invariant
  - Invariant provides a useful property to show the correctness when the loop terminates

Initialization

- Prior to the first iteration of the loop:
  - *i*=*p*-1 and *j*=*p*
  - No values in between *p* and *i*
  - No values in between *i+1* and *j-1*
  - The first two conditions of loop invariant :
    If *p<= k <= i , then A[k] <= x*
    and **If *i +1 <= k <= j-1, then A[k]>x*** are satisfied trivially
  - The third condition of loop invariant: **If *k=r, then A[k] = x*** is satisfied with the assignment in line 1 (x= A[r]) of the pseudo code

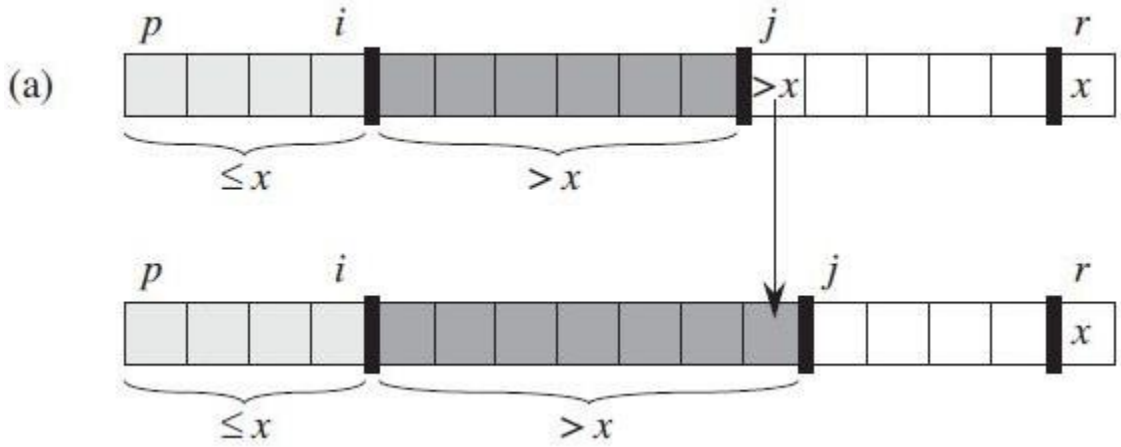
Maintenance

- We consider two cases: (a) *A[j]* >*x* (b) *A[j]* <= *x*



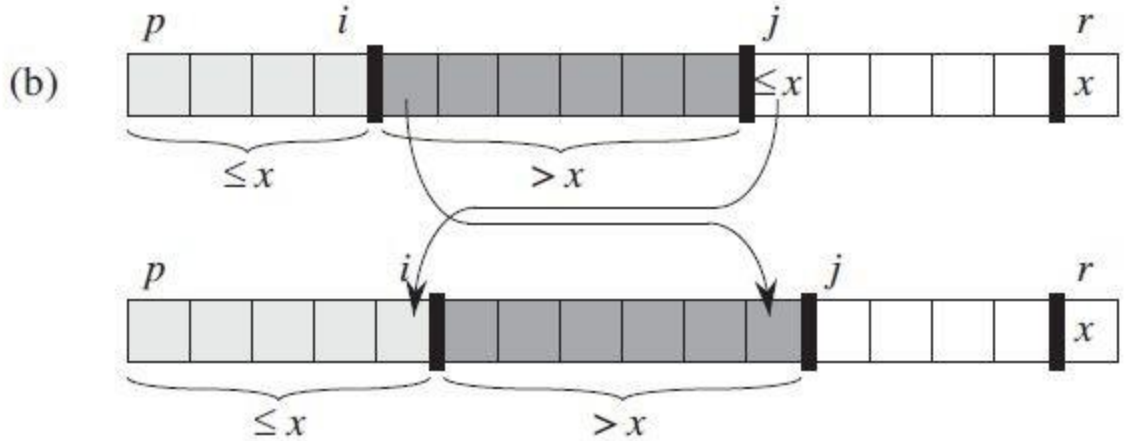
Maintenance : case (a) *A[j] >x*

- Only action in the loop is to increment *j*
- After *j* is incremented, the condition 2 of the loop invariant, **If** ***i +1 <= k <= j-1, then A[k] >x*** holds for *A[j-1]* and all other entries remain unchanged.
- Hence, the loop invariant holds



Maintenance : (b) *A[j] <= x*

- The loop increments *i* , Swaps *A[i]* and *A[j]* & Increments *j*
- Because of swap, *A[i]<=x and* condition 1 of loop invariant **If *p<= k <= i , then A[k] <= x*** is satisfied
- *A[j-1] >x,* the condition 2 of the loop invariant, **If *i +1 <= k <= j-1, then A[k] >x*** holds for A[j-1]



Termination

- At termination *j=r.*
- Every element is in one of the 3 sets described by the loop invariant
- We have partitioned the array into these three sets: those less than or equal to *x,* those greater than *x* and those equal to *x*
- Last two lines of *PARTITION* moves the pivot to its correct place in *A* and returns its index(i+1)

Best case analysis of QUICK sort

- Assume that the pivot p divides the array into balanced i.e. size of A[p..q-1]=A[q+1..r]=n/2
- PARTITION algorithm takes $\Theta(\mathrm{n})$ for tracing the entire array(A) from index p to r-1.
- The recurrence equation is
- $\mathrm{T(n)} = \begin{cases} c & if\ n = 1 \\ 2T\left(\frac{n}{2}\right) + \mathrm{cn} & if\ \mathrm{n} \geq 1 \end{cases}$

By repetitive substitution

$$T(n) = 2T\left(\frac{n}{2}\right) + cn$$

$$T(n) = 2\left[2T\left(\frac{n}{4}\right) + \frac{cn}{2}\right] + cn$$

Cont…

$$T(n) = 2T\left(\frac{n}{2}\right) + cn$$

$$T(n) = 2\left[2T\left(\frac{n}{4}\right) + \frac{cn}{2}\right] + cn$$

$$T(n) = 4T\left(\frac{n}{4}\right) + 2cn$$

Substitute T(n/4) in T(n) then

$$T(n) = 4\left[2T\left(\frac{n}{8}\right) + \frac{cn}{4}\right] + 2cn$$

Simplify it as $T(n) = 8T\left(\frac{n}{8}\right) + 3cn$

Rewrite it as $T(n) = 2^3T\left(\frac{n}{2^3}\right) + 3cn$

Continue the substitutions for k times, then T(n) is

$$T(n) = 2^kT\left(\frac{n}{2^k}\right) + kcn$$

Assume n = 2^k then lg n=k,

$$T(n) = nT\left(\frac{n}{n}\right) + cnlgn \Rightarrow \Theta(nlgn)$$

Worst case analysis of Quick sort

- Assume that the pivot divides the array A[p..r] into unbalanced i.e., A[p..q-1]=n-1, and A[q+1..r]=0, always in each level.
- The recurrence relation is as follows

$$\mathrm{T(n)} = \begin{cases} c & if\ n = 1 \\ T(n-1) + \mathrm{cn} & if\ \mathrm{n} \geq 1 \end{cases}$$

$$T(n) = T(n-1) + \mathrm{cn}$$

After substitution of $\mathrm{T(n-1)}$ into $\mathrm{T(n)}$ is

T(n) = T(n-2)+c(n-1)+cn

After n-1 substitutions the T(n) is

T(n) = T(n-(n-1))+c2+c3+…+c(n-1)+cn

T(n) =c(1+2+…+n-1+n)= cn(n+1)/2=> $\Theta(n^2)$

Thank You